

Olist E-Commerce Business Performance Analysis

A Data-Driven Strategic Review | 2016–2018

Brazilian E-Commerce Public Dataset | Kaggle

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Executive Summary

Between 2016 and 2018, Olist — Brazil's largest e-commerce marketplace — processed approximately 99,000 orders generating around \$16M in Gross Merchandise Value across 27 Brazilian states. The business demonstrated exceptional top-line growth, with monthly revenue scaling from approximately \$138K in January 2017 to over \$1.1M in January 2018 — a year-over-year increase of more than 700% in a single month.

The central finding: Despite strong growth, Olist's repeat purchase rate stands at just 3.12% — against an industry benchmark of 25–30%. 97 out of every 100 customers never came back. This structural retention problem represents the most significant threat to long-term profitability.

This case study analyzes performance across five dimensions — sales growth, customer behavior, geographic intelligence, operational efficiency, and product economics — and concludes with five strategic recommendations with projected revenue impact of approximately \$5.9M annually.

~\$16M

~99,000

\$161

Total GMV	Total Orders	Avg Order Value
4.09/5	3.12%	~96,000
Avg Review Score	Repeat Rate	Unique Customers

1. Business Context

Olist is not a retailer. It is a marketplace — a platform connecting small and medium-sized Brazilian businesses to customers nationwide. Sellers list products on Olist's platform, customers place orders, and third-party logistics partners handle delivery. Olist earns revenue through seller commissions (approximately 12–15% per transaction) and monthly subscription fees.

The ~\$16M figure in this analysis represents platform GMV — the total value of all transactions processed through Olist. Olist's actual earned revenue is estimated at approximately \$1.9–2.4M over this period. Treating GMV as profit would be fundamentally misleading.

Dataset Overview

Dataset	Records	Key Information
Orders	~99,000	Order dates, status, delivery timelines
Order Items	~112,000	Products, prices, freight costs
Customers	~99,000	Location, unique customer identity
Payments	~103,000	Payment method, installments, values
Reviews	~99,000	Customer satisfaction scores (1–5)
Products	~33,000	Categories, dimensions, weight

Seven tables were joined using a star schema data model in Microsoft Power BI, with the orders table as the central fact table.

2. Sales Performance Analysis

Revenue Growth

Period	Monthly Revenue	Notes
January 2017	~\$138K	Early platform stage
November 2017	~\$1.19M	Peak month — Black Friday effect
December 2017	~\$878K	Post-holiday dip
January 2018	~\$1.1M	700%+ YoY growth vs Jan 2017

The November 2017 peak of approximately \$1.19M was driven by Brazil's Black Friday event. The December dip to around \$878K is a natural post-holiday correction. What is most significant is January 2018 at over \$1.1M — nearly matching the November peak without any seasonal driver. By early 2018, Olist had structurally elevated its monthly revenue baseline.

Top Revenue Categories

Rank	Category	Approx Revenue
1	Health & Beauty	~\$1.25M
2	Watches & Gifts	~\$1.2M
3	Bed, Bath & Table	~\$1.04M
4	Sports & Leisure	~\$950K
5	Computer Accessories	~\$880K
10	Tools & Garden	~\$485K

Health & Beauty leads at approximately \$1.25M. The gap between first and tenth place (2.6x) shows meaningful category concentration but not extreme dominance. Computer accessories, while ranking 5th in total revenue, command the highest average order value at over \$1,000 per transaction — revealing two distinct customer profiles on the platform.

Order Fulfillment

Of approximately 99,000 orders placed, around 96,500 (96.48%) were successfully delivered. The roughly 630 canceled orders at \$161 average order value represent approximately \$100K in lost GMV — translating to around \$12–15K in lost Olist commission revenue.

3. Geographic Analysis

State	Approx Revenue	% of Total
São Paulo (SP)	~\$6M	~37%
Rio de Janeiro (RJ)	~\$2.1M	~13%
Other 8 states	~\$8M	~50%

São Paulo generates nearly 3x more revenue than second-place Rio de Janeiro. At city level, São Paulo city alone drives approximately \$2.2M — nearly 14% of total GMV. The top 2 cities combined exceed \$3.3M — more than 20% of all platform revenue.

Geographic risk: Olist's dependence on São Paulo for ~37% of revenue creates real vulnerability. Economic disruption or competitive entry specifically targeting SP could have outsized platform-wide impact.

4. Customer & Payment Analysis

Payment Behavior

Around 78% of all transactions were completed by credit card. Boleto accounted for approximately 18%. The average number of payment installments was approximately 2.6 — stable across all three years with no significant upward trend.

Cash flow implication: At 2.6 average installments, most customers split payments across approximately 3 months. Olist ships the product in month 1 but collects full payment by month 3 — a consistent 2–3 month cash flow lag that becomes increasingly significant as the platform scales.

The Retention Crisis

Repeat purchase rate: 3.12% — This is the most critical finding in the entire analysis.

Of approximately 96,000 unique customers acquired across 2 years, around 93,000 (96.88%) placed exactly one order and never returned. Monthly repeat orders averaged approximately 300 per month throughout the entire dataset period — flat — while new customer orders grew from ~800 in January 2017 to over 7,000 in January 2018.

Benchmark	Repeat Rate
Top performers (Amazon, Flipkart)	40–50%
Average e-commerce marketplace	25–30%
Olist	3.12%

5. Operations Analysis

Delivery Performance by State

State	Delivery vs Estimate	Avg Review Score
PR (Paraná)	~42 days early	4.18 ★
SP (São Paulo)	~41 days early	4.17 ★
BA (Bahia)	~33 days early	3.86 ★ (lowest)
ES (Espírito Santo)	~21 days late	4.04 ★

The Delivery-Satisfaction Paradox

The most counterintuitive finding: Bahia receives orders ~33 days ahead of schedule — yet records the lowest satisfaction score at 3.86. Espírito Santo delivers 21 days late — yet scores a reasonable 4.04. This definitively disproves the hypothesis that delivery speed drives customer satisfaction on Olist.

Conclusion: Product quality and accurate product representation are stronger drivers of customer satisfaction than delivery timing. Olist's improvement strategy should prioritize

seller quality standards over logistics optimization.

6. Product Economics Analysis

Freight Cost as % of Product Price

Category	Freight/Price Ratio	Implication
Casa Conforto (Home Comfort)	~53%	Critical — conversion killer
Flores (Flowers)	~44%	High — perishable handling
Moveis Colchao (Mattresses)	~38%	Expected for large items
PCs (Computers)	~4.4%	Excellent — high value product

A ~53% freight-to-price ratio for home comfort products means customers pay approximately 53 cents in shipping for every dollar of product. Analysis confirmed a positive correlation between product weight and freight cost — categories above 8,000g average weight show freight costs roughly double the platform average.

7. Strategic Recommendations & Projected Outcomes

1 Customer Retention Program • Critical Priority

Actions: Post-purchase email sequence, basic loyalty points program, personalized re-engagement for top 10% spenders.

Projected outcome: Moving repeat rate from ~3% to 8% = ~7,700 additional repeat customers × \$161 = ~\$1.24M incremental GMV → ~\$185K additional Olist commission annually.

2 Geographic Diversification • High Priority

Actions: Recruit sellers in MG, RS, PR. Reduced commission rates for 6 months in underrepresented states. Targeted campaigns in Belo Horizonte, Porto Alegre, Curitiba.

Projected outcome: If MG, RS, PR each reach RJ's ~\$2.1M revenue level → ~\$3.5M incremental GMV → ~\$525K additional Olist commission annually.

3 **Freight Cost Optimization** • Medium Priority

Actions: Negotiate category-specific logistics rates. Freight subsidy for products under \$15 with ratio above 40%. Free shipping threshold at \$50+ basket.

Projected outcome: Conservative 15% reduction in cart abandonment for high-ratio categories → ~\$360K incremental GMV → ~\$54K additional Olist commission annually.

4 **Seller Quality Program** • Medium Priority

Actions: Seller quality scoring based on review scores. Mandatory listing quality training for sellers below 4.0. Surface seller scores during product discovery.

Projected outcome: 0.2 improvement in avg review score → ~3–5% conversion improvement → ~\$800K incremental GMV → ~\$120K additional commission annually.

5 **Espírito Santo Logistics Audit** • Medium Priority

Actions: Audit the specific logistics partner serving ES. If SLA performance is below contract terms, escalate or replace. ES is the only state with consistent late delivery — almost certainly a partner-specific issue.

Combined Impact Summary

Recommendation	Incremental GMV	Olist Revenue
Retention Program	~\$1.24M	~\$185K
Geographic Diversification	~\$3.5M	~\$525K
Freight Optimization	~\$360K	~\$54K
Seller Quality Program	~\$800K	~\$120K
ES Logistics Audit	Modest	Immediate

Recommendation	Incremental GMV	Olist Revenue
Total	~\$5.9M	~\$884K

8. Data Limitations & Methodology

Dataset covers September 2016 to August 2018 — post-August figures in monthly charts reflect 2017 data only. The ~\$16M figure represents GMV not Olist's actual earned revenue. Repeat rate was calculated using customer_unique_id — not customer_id which is unique per order. Freight analysis used price from order_items at product level, not payment_value at order level. All analysis performed in Microsoft Power BI Desktop with six custom DAX measures across a star schema data model.

9. Conclusion

Olist demonstrated exceptional growth between 2016 and 2018 — monthly revenue grew 700%+ year-on-year, order volume scaled nearly 10x, and ~96.5% of orders were successfully delivered with a 4.09/5 satisfaction score. The operational foundation is genuinely solid.

But the business is running a leaking bucket. It acquires customers efficiently and delivers products reliably — then loses 97% of those customers permanently. The five recommendations in this analysis — with a combined projected GMV impact of approximately \$5.9M annually — represent a clear, data-backed path from growth to sustainable profitability.

The retention program is the highest priority. Everything else builds on a returning customer base.